Dental Multi-Purpose Clinic and Education Platform — Implementation Plan

1. Goal and product summary

Build a polished, mobile-friendly **multi-tenant website and booking platform** for dental clinics, education centers, and practical training facilities. The product should not be tied to one clinic name, logo, or visual identity. Each participating clinic should be able to configure its own profile, branding, locations, resources, services, prices, policies, and operating hours, while students and patients discover services, reserve resources, pay required deposits online, and receive clear booking confirmations.

The first release will include a public customer marketplace/discovery experience, clinic profiles, authenticated user accounts, live availability booking, online deposit payments in Egyptian pounds through **Paymob**, an installable desktop-style operations application for clinic staff, and the tooth-selection assistance workflow. The system must prevent double bookings even when staff create manual reservations or block times after a phone call. The operations app should be installable from the browser as a desktop-style **Progressive Web App (PWA)**, with a future option to package the same admin interface as a native desktop wrapper if the clinics later require deeper operating-system integration.

The terminology used throughout the site should be professional and medically appropriate. The service the user described as cleaning dental tools should be presented as **instrument sterilization and reprocessing** or **dental instrument sterilization**, subject to confirmation from the clinic.

2. Decisions already made

Area	Decision
Payments	Users pay a deposit online before a booking is confirmed. The remaining balance, if any, can be configured as payable at the clinic.
Booking	Availability is live and bookings are confirmed automatically when payment succeeds.

Staff control	Admins can manually block unavailable hours and can create bookings for phone or walk-in requests.
First release	Include chair rental, service/course showcases, online deposits, admin tools, cleaning/sterilization timing, and tooth selection.
Main chair price	Start with **120 EGP per hour**, stored as configurable data rather than hard-coded UI text.
Locale	Use Egyptian pounds and the **Africa/Cairo** timezone as the initial operating timezone.
Repository	Work will be applied to the selected GitHub repository `BasantAwad/Dental` after approval and after inspecting its current state.
Platform scope	Multi-clinic from the first release; no clinic-specific name, logo, or hard-coded branding.
Admin application	Installable staff-facing PWA with offline-friendly shell, persistent sign-in where supported, large-calendar workflow, and role-based clinic access.
Payment provider	Paymob, using verified server-side callbacks/webhooks and idempotent payment handling.
Deposit rule	A booking is not confirmed until at least **50% of the total amount** has been successfully paid.
Cancellation rule	If a user cancels **within 24 hours before the booking**, the deposit is not refunded. The exact treatment for cancellations earlier than 24 hours remains configurable and should be confirmed by clinic policy.

3. Recommended product structure

Public website

The public-facing experience should use a neutral, multi-clinic-ready clinical-education visual system rather than a single-clinic brand. The recommended direction is a warm, high-trust palette combining deep navy or charcoal, clinical ivory, and a restrained aqua or mint accent, with generous whitespace, strong typography, subtle depth, and restrained motion. Clinic-specific logos, colors, names, and profile imagery should be injected through tenant configuration rather than embedded in the codebase.

The primary navigation should include **Explore Clinics**, **Services**, **Chair & Dummy Head Rental**, **Courses**, **Tooth Assistance**, and **Help**. A persistent call to action should lead users to booking. On mobile, the navigation should collapse cleanly and retain a visible booking action. A clinic profile should expose its own logo/name, location, hours, resources, policies, and service catalog without requiring a separate code deployment.

Proposed pages

Page	Purpose and key behavior
Home	Introduce the clinic and training center, show the logo/name, feature an interactive rotating tooth or dental model, explain the three major audiences, highlight services, show chair pricing, and provide booking CTAs.
Services	Present clinical services, radiography, instrument sterilization/reprocessing, dummy-head practice, chair rental, and other configurable services as cards with duration, price/deposit, and booking links.
Chair Rental	Show a chair image or gallery on the left and a live calendar/slot picker on the right. Display 120 EGP/hour initially, selected duration, deposit, remaining amount if applicable, policies, and checkout.
Courses	Show educational courses with dates, instructors, capacity, syllabus summary, pricing/deposit, and enrollment. Course enrollment should use the same booking/payment primitives where possible.
Tooth Assistance	Let a student choose **Upper**, **Lower**, or **Both** arches, then select teeth on an interactive model or fallback image, add a short assistance request, and submit it for a consultation/assistance booking.

Booking result/account	Display payment status, booking reference, date/time, resource, deposit, balance, cancellation policy, and calendar-friendly details. Logged-in users can view upcoming and past bookings.
About/Contact	Explain the clinic, doctors, teaching approach, location, hours, contact channels, and policies without exposing unnecessary medical information.

4. Tooth-selection experience

The preferred experience is an interactive 3D dental model for the upper and lower arches. Users should be able to rotate, zoom, reset the view, and click individual teeth to highlight them. Selected teeth should have a clear visual state and should also be represented in an accessible list or odontogram-style summary so the selection is not dependent on color alone.

The homepage model should be a lightweight decorative/educational 3D experience, while the Tooth Assistance page should use a more functional selectable model. The referenced Sketchfab models by Hayashiya Samboku may be evaluated as candidate assets, but their **license, embedding terms, download permissions, attribution requirements, file format, performance, and suitability for per-tooth hit detection must be verified before use**. If the model cannot support reliable individual tooth selection, the implementation should use a prepared GLB/GLTF asset with separate tooth meshes or a deterministic 2D fallback.

The fallback plan is an annotated full-open-mouth image with tooth regions mapped to selectable overlays. It should preserve the same API and stored representation as the 3D version: arch, tooth identifier, display label, and optional surface/notes. This means the business workflow remains usable on devices that cannot render WebGL or when the 3D asset is unavailable.

The request form should collect only what is necessary: selected arch/teeth, optional notes, preferred appointment type, and contact details available from the user account. It should avoid collecting identifiable patient clinical records unless the clinic explicitly requires them and has an approved privacy process.

5. Booking and availability architecture

Live availability is the most important reliability feature. Availability should be calculated per clinic, location, and resource from a single source of truth combining recurring operating hours, existing confirmed/pending bookings, admin-created phone bookings, one-off closures, maintenance periods, and manual blackout blocks. Every reservation must carry a tenant/clinic ID so one clinic can never affect another clinic’ s inventory.

All booking times should be normalized to the relevant clinic timezone and stored with unambiguous timestamps. The server, not the browser, must validate availability immediately before creating a hold or booking. The booking flow should use a short-lived pending hold during Paymob checkout, then convert it to a confirmed booking only after the verified Paymob callback reports that at least 50% of the total has been paid. Expired holds must release their slots automatically.

The database must enforce conflict prevention in addition to UI checks. The implementation should use a transaction and a conflict-safe reservation strategy so two users attempting the same chair/hour cannot both succeed. Admin-created bookings and blackout blocks must go through the same conflict validation path as public bookings. Admins should see a clear conflict message and the currently occupied slot rather than silently overwriting it. The calendar must also calculate a resource’ s **day availability state**: hourly reservations display their exact occupied intervals, while the entire day is labeled **Fully reserved** only when no bookable slot remains after considering operating hours, buffers, existing reservations, and blocks. A partially occupied day should show **Partially booked**, and a day with no remaining eligible slots should be disabled for that resource. This applies to both dental chairs and dummy heads.

The initial resource model should support multiple resource types across multiple clinics and locations: dental chair, dummy head, x-ray room/device, and sterilization/reprocessing service. Each resource should have its own availability, duration rules, buffer time, price, deposit rule, and active/inactive status. A clinic administrator should be able to configure whether a resource is bookable hourly, by a fixed session, or as an all-day reservation where appropriate.

6. Payment flow

Use a payment-provider adapter while implementing **Paymob as the first provider**. Each clinic should have a payment configuration/status record so the platform can support multiple clinic accounts or a centrally managed merchant arrangement without changing booking logic. The integration should support online EGP deposits, payment initiation, success/failure/cancellation states, verified server-side callbacks/webhooks, idempotency, and reconciliation in the admin panel. Paymob credentials must remain server-side and must never be exposed to the browser.

A booking should never be considered paid based only on a client-side redirect. The server should verify the Paymob result, record the provider transaction reference, mark the deposit as paid exactly once, and confirm the booking only when the paid amount is at least 50% of the total. If payment fails or expires, the temporary slot hold should be released. The admin panel should show payment status and allow staff to record offline balances without altering the immutable online payment record. Cancellation/refund logic must calculate the 24-hour cutoff using the clinic’ s timezone and must preserve the original payment and refund decision in the audit trail.

Prices should be stored as integer minor units to avoid floating-point errors. Deposit percentages or fixed deposit amounts should be configurable per service. The checkout summary must show the selected date/time, resource, duration, gross price, deposit due now, remaining amount, and cancellation/refund policy before the user pays.

7. Admin dashboard

The admin side should use a persistent sidebar layout with role-aware access control and be delivered as an installable **staff operations PWA**. Staff should be able to install it to the desktop, launch it from an application icon, and work in a focused operations interface rather than keeping the public website open. The app should provide a responsive desktop-first layout, clear connection state, safe retry behavior, and an offline shell for navigation; booking writes and payment operations must require a live connection and must never be falsely shown as completed offline. At minimum, define platform support, clinic owner/administrator, manager, and staff permissions separately from normal users. Clinic owners should only access their own tenant’ s data unless explicitly granted platform-level access.

Admin area	Required capabilities
Overview	Today’ s bookings, deposits collected, occupancy, upcoming courses, pending assistance requests, blocked hours, operational alerts, and installation/connection status for the staff app.
Calendar	Day/week views, filters by resource and status, booking details, manual booking creation, rescheduling/cancellation, and conflict-safe block creation.
Availability	Operating hours, holidays, one-off closures, recurring unavailable times, maintenance, manual phone-call blocks, and a calendar

	legend for hourly occupied, partially booked, fully reserved, unavailable, and pending-hold states.
Services and resources	Create/edit services, prices, deposits, durations, buffers, images, descriptions, capacity, and resource assignment.
Courses	Manage course descriptions, instructors, dates, capacity, enrollment status, pricing, and attendance list.
Tooth assistance	Review submitted tooth selections, notes, requested service, status, and follow-up appointment.
Sterilization/reprocessing	Configure turnaround times, receive service bookings, track status, and see cleaning/reprocessing workload by day and period.
Payments	Search by booking/reference, view payment status, provider transaction ID, refunds/adjustments if supported, and export a reconciliation report.
Reports	Revenue/deposits, utilization by chair/resource, no-shows/cancellations, popular services, course enrollment, and sterilization workload/timing statistics.
Settings	Clinic profile/branding, location, timezone, policies, contact information, deposit rules, cancellation cutoff, Paymob configuration/status, notification templates, staff roles, and resource-level booking rules.

For sterilization, the system should distinguish **booking time** from **reprocessing turnaround time**. A service can define a duration and a required buffer, and the admin calendar should show both the customer-facing appointment and the operational cleaning/reprocessing window. This prevents a short booking from being treated as immediately available when staff need additional time.

8. Core data model

The schema should be designed before UI implementation and should keep operational records auditable. A likely first schema includes a platform/tenant boundary from the

beginning:

Entity	Important fields
PlatformUser / User	Auth identity, platform role, name, phone/email, created/updated timestamps.
Clinic	Name, slug, logo, brand settings, contact details, timezone, active state, policy settings.
ClinicMembership	User, clinic, role, invitation/status metadata, created/updated timestamps.
Service	Name, category, description, active state, duration, price, deposit rule, required buffer, resource type.
Resource	Name, type, location, active state, capacity, booking rules.
OperatingHours	Resource or clinic scope, weekday, start/end time, effective dates.
AvailabilityBlock	Clinic/resource, start/end, reason, created by, source such as phone call/maintenance/holiday, and whether it is recurring.
Booking	User, service, resource, start/end, status, source, notes, deposit due, balance due, cancellation data, reference.
BookingHold	Booking/session identifier, resource, start/end, expiration time, checkout status.
Payment	Booking, clinic, provider (`paymob`), provider transaction ID, amount, currency, status, verified timestamp, raw-safe metadata, idempotency key.
Course	Title, instructor, description, dates, capacity, price/deposit, enrollment state.
ToothAssistanceRequest	User, booking/course linkage, arch choice, selected teeth, notes, status, assigned staff member.
ToothModelAsset	Arch, asset URL, version, interaction mode, attribution/license metadata, active state.

SterilizationJob or ReprocessingWindow	Booking/service linkage, start/end, status, assigned staff, completion timestamp, notes.
Notification	Recipient, event type, channel, delivery status, sent timestamp.
AuditLog	Actor, action, entity, entity ID, timestamp, safe before/after summary.

The selected tooth identifiers should use a consistent notation agreed with the clinic, preferably FDI numbering for an Egyptian dental audience, with a display-label configuration so the interface can explain the notation. The schema should allow multiple selected teeth and should not store raw medical records by default.

9. Technical approach

Use the existing full-stack project architecture in the repository after inspection: React and Tailwind for the interface, typed server procedures for all application calls, a relational database schema for bookings and payments, and the built-in authentication flow for user accounts. Reuse the existing dashboard layout and UI primitives where present rather than introducing a second design system.

The frontend should be divided into public pages, authenticated user pages, and admin pages. Backend procedures should be separated by domain: services/resources, availability, bookings, payments, courses, tooth assistance, sterilization, notifications, reports, and admin settings. Every procedure should enforce authorization server-side.

Large images and 3D assets should be stored through the project’ s managed storage workflow rather than committed into the frontend bundle. The initial 3D experience should be lazy-loaded, show a static poster while loading, provide a reduced-motion option, and fail gracefully to the 2D mouth-image mode on unsupported devices or slow connections.

10. Two viable implementation options for the interactive tooth model

Approach	Tradeoffs	Cost	Setup complexity
Interactive 3D primary with 2D fallback	Best matches the vision and creates a strong differentiator. Requires properly prepared/licensed assets, WebGL	Medium to high asset/integration effort; payment and hosting costs depend on chosen providers.	High

	performance work, per-tooth mesh mapping, and a fallback for low-end/mobile devices.		
2D selectable mouth image first, 3D as a later enhancement	Faster, more predictable, accessible, and easier to validate with students. Less immersive and does not fully deliver the original 360-model vision at launch.	Low to medium	Low to medium

The plan is to build the shared tooth-selection data contract and UI abstraction first, then make the 3D model the primary renderer only after asset licensing and per-tooth interaction are validated. If the 3D asset fails that validation, the 2D experience becomes the production renderer without redesigning the rest of the workflow.

11. Delivery phases

Phase 0 — Repository and requirements validation

Inspect the current `BasantAwad/Dental` repository, identify the existing framework, routes, schema, authentication state, and reusable components, and preserve working project conventions. Confirm the platform owner's marketplace rules, final service taxonomy, clinic onboarding fields, operating-hours rules, staff roles, Paymob account/configuration model, and preferred tooth-numbering notation. Verify the Sketchfab asset license and technical suitability before implementation.

Phase 1 — Multi-clinic foundation and public experience

Establish the tenant-aware data model and visual system, responsive public shell, clinic discovery/profile pages, services catalog, courses catalog, contact/about pages, user authentication entry points, and managed media assets. Build the interactive homepage/model experience as a progressive enhancement with a static fallback. Add the installable admin PWA shell, manifest, icons, install prompt guidance, desktop-first navigation, and connection-state UI.

Phase 2 — Database and booking engine

Implement the relational schema, service/resource configuration, operating hours, blackout blocks, booking holds, conflict-safe reservation transactions, cancellations, and user booking history. Add server-side validation and automated tests for overlapping bookings, expired holds, admin blocks, timezone boundaries, and resource-specific availability.

Phase 3 — Paymob deposit checkout

Add Paymob integration, EGP pricing, the mandatory 50% deposit calculation, checkout summary, server-side callback/webhook verification, idempotent payment processing, booking confirmation, failure recovery, 24-hour cancellation/refund decisioning, and payment audit records. Add confirmation notifications using the project’ s supported notification path after the payment state is reliable.

Phase 4 — Admin operations and reporting

Build the admin calendar, manual booking/block workflows, service/resource settings, course management, sterilization/reprocessing timing view, user and booking search, payment reconciliation, and operational statistics. Add role checks, audit logging, and exportable reports where appropriate.

Phase 5 — Tooth assistance

Implement Upper/Lower/Both selection, 3D or 2D model renderer abstraction, tooth highlighting, accessible selected-tooth list, FDI labels, notes, request status, and admin review. Connect the request to an assistance booking or follow-up workflow without exposing private requests publicly.

Phase 6 — Desktop operations, hardening, and launch readiness

Test the customer website and installable admin PWA on mobile and desktop, validate Paymob callbacks in a sandbox, verify race-condition behavior with concurrent booking attempts, test full-day calendar states, test PWA installation and reconnect behavior, test WebGL fallback, review accessibility, optimize image/model loading, seed only non-sensitive demo data, and prepare production configuration and multi-clinic onboarding documentation. Keep native desktop packaging as a later enhancement unless a clinic specifically requires OS-level features.

12. Test and acceptance plan

The booking engine is accepted only if two simultaneous users cannot obtain the same chair and time, and if an admin block immediately removes that time from public availability. Tests should cover hourly rental at 120 EGP as configurable seed data, multi-

hour selections, pending payment expiry, failed Paymob payments, successful confirmation only at 50% or more paid, cancellation within and beyond the 24-hour cutoff, course capacity limits, dummy-head and x-ray resource separation, sterilization buffers, tenant isolation, and admin-created phone bookings.

The admin workflow is accepted only if staff can install and launch the PWA, create phone bookings, manually block hours, see the reason and actor, identify conflicts before saving, and view the effect on the public calendar. A day with some occupied slots must remain partially bookable and show **Partially booked**; a day with no remaining eligible slots must show **Fully reserved** and prevent new selection. Reports should reconcile against booking and payment records and should not double-count retries or duplicate callbacks.

The tooth workflow is accepted only if a user can select Upper, Lower, or Both, select multiple teeth, see a persistent text summary of the selected teeth, submit notes, and allow an admin to review the request. The same request must work when the 3D renderer is disabled and the 2D fallback is used.

The public website is accepted only if it is responsive, keyboard-navigable, readable with reduced motion, has loading/empty/error states, uses accessible labels for model controls, and does not rely on color alone to communicate tooth selection or booking status.

13. Assumptions and open risks

The plan assumes the platform owner will provide the marketplace’ s legal/business details and that each participating clinic will enter its own legal/business name, logo, contact information, operating hours, service descriptions, pricing for services beyond chair rental, cancellation/refund rules, and staff accounts through onboarding. Paymob is the chosen initial payment provider; implementation still requires the platform owner to provide the correct Paymob merchant, integration, iframe/checkout, callback/webhook, and environment configuration. The adapter should retain a provider boundary so another gateway can be added later without rewriting booking logic.

The plan assumes each participating clinic is permitted to collect the selected tooth information for assistance scheduling and will define its privacy/consent requirements. The platform owner must also decide whether customers select a clinic first or can compare multiple clinics before booking; the plan assumes clinic discovery first, followed by booking within one clinic/resource context. The first release should minimize health data collection and should not become a full electronic medical record system unless that is separately requested and designed with appropriate compliance controls. Tenant isolation, clinic-level authorization, and auditability are mandatory because multiple independent clinics will use the same platform.

The largest technical risk is the availability and suitability of the referenced 3D models. Licensing, file size, separate tooth meshes, mobile performance, and reliable picking must

be validated. The 2D image renderer is therefore a required fallback, not an optional afterthought.

A second risk is operational ambiguity around whether a chair booking, doctor assistance session, x-ray, dummy-head practice, and instrument reprocessing can overlap. Each should be represented as a separate resource/service rule so the admin can configure reality without code changes.